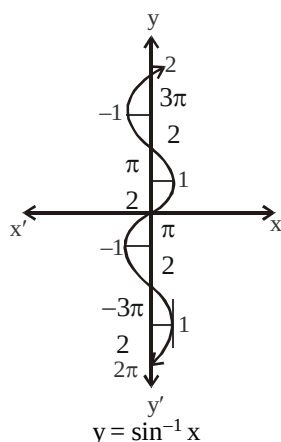


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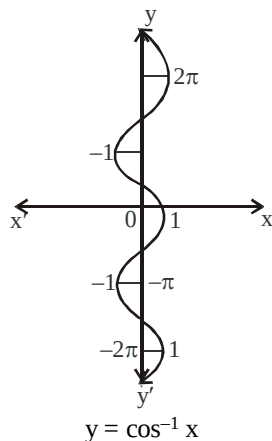
INVERSE TRIGONOMETRIC FUNCTIONS

KEY CONCEPT INVOLVED

- | Functions | Domain | Range |
|-----------------------------|--|------------------------|
| (i) $\sin$ | $\mathbb{R}$ | $[-1, 1]$ |
| (ii) $\cos$ | $\mathbb{R}$ | $[-1, 1]$ |
| (iii) $\tan$ | $\mathbb{R} - \{x : x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}\}$ | $\mathbb{R}$ |
| (iv) $\cot$ | $\mathbb{R} - \{x : x = n\pi, n \in \mathbb{Z}\}$ | $\mathbb{R}$ |
| (v) $\sec$ | $\mathbb{R} - \{x : x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}\}$ | $\mathbb{R} - [-1, 1]$ |
| (vi) cosec | $\mathbb{R} - \{x : x = n\pi, n \in \mathbb{Z}\}$ | $\mathbb{R} - [-1, 1]$ |
- Inverse Function** - If $f : X \rightarrow Y$ such that $y = f(x)$ is one-one and onto, then we define another function $g : Y \rightarrow X$ such that $x = g(y)$, where $x \in X$ and $y \in Y$ which is also one-one and onto. In such a case domain of g = Range of f and Range of g = domain of f
 g is called inverse of f or $g = f^{-1}$
 Inverse of $g = g^{-1} = (f^{-1})^{-1} = f$.
- Principal value Branch of function $\sin^{-1}$** - It may be noted that for the domain $[-1, 1]$ the range could be any one of the intervals $\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right], \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ or $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ corresponding to each interval we get a branch of the function $\sin^{-1}$ the branch with range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is called the principal value branch.
 Thus $\sin^{-1} : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

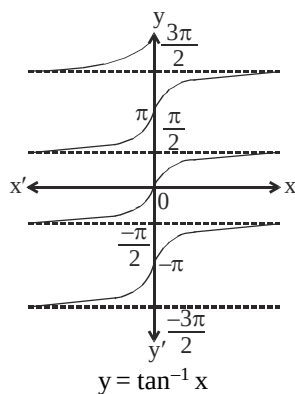


4. **Principal Value branch of function $\cos^{-1}$** - Domain of the function $\cos^{-1}$ is $[-1, 1]$. Its range is one of the intervals $(-\pi, 0)$, $(0, \pi)$, $(\pi, 2\pi)$. etc. The branch with range $(0, \pi)$ is called the principal value branch of the function $\cos^{-1}$ thus $\cos^{-1} : [-1, 1] \rightarrow [0, \pi]$



5. **Principal Value branch of function $\tan^{-1}$** - The function $\tan^{-1}$ is defined whose domain is set of real numbers and range is one of the intervals $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right)$, $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ etc.

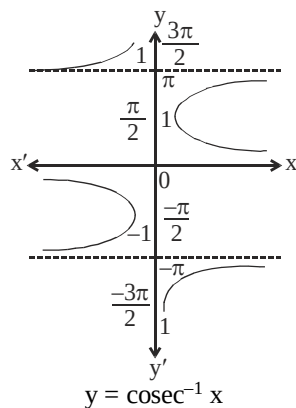
Graph of the function is as shown in the adjoining figure the branch with range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is called the principal value branch of function $\tan^{-1}$. Thus $\tan^{-1} : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.



6. **Principal Value branch of function $\operatorname{cosec}^{-1}$** - The function $\operatorname{cosec}^{-1}$ is defined on a function whose domain is $\mathbb{R} - (-1, 1)$ and the range is anyone of the interval $\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right] - \{\pi\}$, $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$, $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right] - \{\pi\}$,

The function corresponding to the range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$ is called the principal value branch of $\operatorname{cosec}^{-1}$

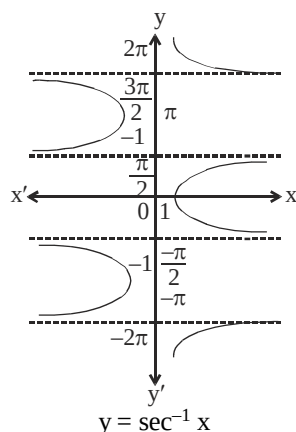
Thus, $\operatorname{cosec}^{-1} : \mathbb{R} - (-1, 1) \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$



$$y = \operatorname{cosec}^{-1} x$$

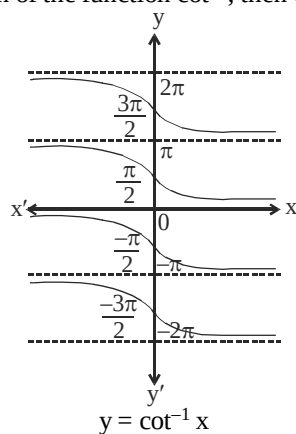
7. Principal value branch of function $\sec^{-1}$ - The $\sec^{-1}$ is defined as a function whose domain is $\mathbb{R} - (-1, 1)$ and the range could be any of the intervals is, $[-p, 0] - \left\{\frac{-\pi}{2}\right\}, [0, p] - \left\{\frac{\pi}{2}\right\}, [\pi, 2\pi] - \left\{\frac{3\pi}{2}\right\}$ etc.

The branch corresponding to range $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ is known as the principal value branch of $\sec^{-1}$. Thus $\sec^{-1} : \mathbb{R} - (-1, 1) \rightarrow [0, \pi] - \left\{\frac{\pi}{2}\right\}$.



$$y = \sec^{-1} x$$

8. Principal Value branch of function $\cot^{-1}$ - The $\cot^{-1}$ function is defined as the function whose domain is $\mathbb{R}$ and the range is any of the intervals..... $(-\pi, 0), (0, \pi), (\pi, 2\pi)$ etc. The branch corresponding to $(0, \pi)$ is called the principal value branch of the function $\cot^{-1}$, then $\cot^{-1} : \mathbb{R} \rightarrow (0, \pi)$



$$y = \cot^{-1} x$$

9.	Inverse function	Domain	Principal Value branch
	$\sin^{-1}$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
	$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
	$\operatorname{cosec}^{-1}$	$\mathbb{R} - (-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
	$\sec^{-1}$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
	$\tan^{-1}$	$\mathbb{R}$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
	$\cot^{-1}$	$\mathbb{R}$	$(0, \pi)$

CONNECTING CONCEPTS

- $\sin^{-1} 1/x = \operatorname{cosec}^{-1} x, x \geq 1, x \leq -1$
 - $\cos^{-1} 1/x = \sec^{-1} x, x \geq 1, x \leq -1$
 - $\tan^{-1} 1/x = \cot^{-1} x, x > 0$
 - $\operatorname{cosec}^{-1} 1/x = \sin^{-1} x, x \in [-1, 1]$
 - $\sec^{-1} 1/x = \cos^{-1} x, x \in [-1, 1]$
 - $\cot^{-1} 1/x = \tan^{-1} x, x > 0$
- $\sin^{-1}(-x) = -\sin^{-1} x, x \in [-1, 1]$
 - $\tan^{-1}(-x) = -\tan^{-1} x, x \in \mathbb{R}$
 - $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x, |x| \geq 1$
 - $\cos^{-1}(-x) = \pi - \cos^{-1} x, x \in [-1, 1]$
 - $\sec^{-1}(-x) = \pi - \sec^{-1} x, |x| \geq 1$
 - $\cot^{-1}(-x) = \pi - \cot^{-1} x, x \in \mathbb{R}$
- $\sin^{-1} x + \cos^{-1} x = \pi/2, x \in [-1, 1]$
 - $\tan^{-1} x + \cot^{-1} x = \pi/2, x \in \mathbb{R}$
 - $\operatorname{cosec}^{-1} x + \sec^{-1} x = \pi/2, |x| \geq 1$
- $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}, xy < 1$
 - $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}, xy > -1$
 - $2 \sin^{-1} x = \sin^{-1} \left(2x\sqrt{1-x^2} \right), -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$
 - $2 \cos^{-1} x = \cos^{-1} (2x^2 - 1), -\frac{1}{\sqrt{2}} \leq x \leq 1$
 - $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2}, -1 < x < 1 = \sin^{-1} \frac{2x}{1+x^2}, |x| \leq 1 = \cos^{-1} \frac{1-x^2}{1+x^2}, x \geq 0$